

Problem 1

Assume a program accepts inputs only if they start with `DATA:`.

- Using a 100-character set, what is the probability that a random 5-character string begins with `DATA:`?
- Starting from `DATA:123`, why does applying one `flip_random_character` mutation make it much more likely that the result still starts with `DATA:` compared to random generation?

Problem 2

Assume you have a seed: `FUZZ`

You perform the following mutations:

- insert `'1'` at index 2 result in `FU1ZZ`
- delete index 0 resulting in `U1ZZ`
- flip a bit in the last character where one possibility is `U1Z[`

Give one possible final string using the mutations above in sequence (e.g., `U1Z[`) and explain how adding such mutated inputs back to the seed pool increases diversity (more varied strings leads to broader exploration of program behavior).

Problem 3

You want to configure a `fuzzer` to only generate uppercase English letters (A-Z). The ASCII code for `'A'` is 65. What values should you provide for `char_start` and `char_range` in the `fuzzer` call to achieve this?

Problem 4

A C program uses `atoi()` on fuzzed input to allocate memory:

```
char *buf = malloc(atoi(fuzzed_string));
```

Give two input examples from `fuzzer()` that could cause a crash or hang along with the arguments for the call that triggers this.

(We do not assume you know the specification of `atoi` and `malloc`. If such a question appears in the test, we will explain the specification for both these calls. For the given problem, you can perform a quick search to get the spec.)